PLS Discriminant Analysis

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Introduction

PLS discriminant analysis (PLS-DA) extends partial least squares regression to classification by dummy-coding the class variable as a binary indicator matrix. It has become a standard tool in metabolomics, transcriptomics, and other omics disciplines because it handles the $p \gg n$ regime gracefully and, in its sparse variant, performs simultaneous classification and feature selection. The output — a small set of latent components plus the discriminant boundary in the projected space — is interpretable in a way that black-box classifiers (random forests, deep nets) often are not.

Prerequisites

A working understanding of PLS regression, classification metrics, and the high-dimensional ($p \gg n$) regime that motivates latent-component methods.

Theory

Encode the class variable as one-hot indicators $Y \in \{0, 1\}^{n \times K}$ for K classes. PLS regression on (X, Y) produces latent components that maximise covariance with the indicator matrix; the predicted indicators are continuous, and a new sample is assigned to the class with the largest prediction. The number of components is selected by cross-validation (`mixOmics::perf()`).

Sparse PLS-DA (sPLS-DA) imposes an L1 penalty per component, retaining only `keepX` predictors per component. This produces a signature — a small list of variables driving the classification — that can be reported alongside the classifier itself.

Assumptions

Linear relationships between latent components and outcome probabilities, sample sizes large enough per class to estimate the components stably (rule of thumb: at least 5–10 observations per class), and meaningful covariance between predictors and the encoded indicators.

R Implementation

```r
library(mixOmics)

data(srbct)
X <- srbct$gene
Y <- srbct$class

# PLS-DA
fit_plsda <- plsda(X, Y, ncomp = 3)
plotIndiv(fit_plsda, ind.names = FALSE, legend = TRUE)

# Sparse PLS-DA: select 50 variables per component
fit_splsda <- splsda(X, Y, ncomp = 3, keepX = c(50, 50, 50))
selectVar(fit_splsda, comp = 1)$name |> head()
```

Output & Results

`plsda()` returns latent component scores per sample, loadings for each predictor, and predicted class labels. `splsda()` adds variable-selection-per-component, with `selectVar()` extracting the chosen predictors. `plotIndiv()` displays samples in the latent component space, coloured by class.

Interpretation

A reporting sentence: "Sparse PLS-DA on the 2,308-gene SRBCT dataset selected 50 genes per component across three components, achieving 100 % balanced accuracy in 10-fold cross-validation; the resulting 150-gene signature substantially overlaps with the published transcriptomic markers of the four cancer subtypes." Always report both classification performance (accuracy, balanced accuracy, AUC) and the selected feature signature.

Practical Tips

- Use `mixOmics::perf()` with cross-validation to select `ncomp` and `keepX`; a fixed choice without tuning over-fits in high-dimensional omics.
- For two-class problems, logistic regression with elastic net regularisation often performs comparably and is more familiar; reserve sPLS-DA for multi-class or when the covariance structure is strongly multivariate.
- Sparse PLS-DA can produce unstable variable selections in the presence of highly correlated predictors; report selection stability via bootstrap resampling (`mixOmics::perf()` with `nrepeat = 50`).
- The selected signature is the headline result for biological audiences; pair it with pathway-enrichment analysis to interpret functionally.
- Compare against alternative classifiers (random forest, elastic-net logistic, sPLS-DA) on identical CV folds; modern omics papers typically report multiple methods.
- For very imbalanced classes, use `near.zero.var = TRUE` and balanced cross-validation to prevent the majority class from dominating component construction.

R Packages Used

`mixOmics` for `plsda()`, `splsda()`, `perf()`, `selectVar()`, and the broader multi-block extensions; `caret` for unified cross-validated benchmarking against alternative classifiers; `ropls` for an alternative PLS-DA implementation popular in metabolomics; `glmnet` for elastic-net logistic regression as a complementary high-dimensional classifier.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE